CASE STUDY · RETENTION ANALYTICS

Reducing SMB Cardholder Churn with ML

An end-to-end propensity system — from problem framing to production — that lifted retention by $8–12M annually.

2.5–3.5%	$8–12M	4 wks → <1 wk	97%
CHURN REDUCTION	ANNUAL RETAINED REVENUE	DEPLOYMENT TIME	PIPELINE SUCCESS RATE

Arun · Senior Data Scientist · Mastercard (via Fulcrum Digital)

SMB churn was rising — and our marketing was flying blind.

THE TRIGGER

Marketing flagged rising churn in the small-business cardholder portfolio and asked whether we could move from one-size-fits-all campaigns to targeted, predictive outreach.

WHAT THE DATA TOLD US

A meaningful share of customers who eventually churned had received zero proactive engagement — a clear signal that a predictive, segmented approach could move the needle.

Untargeted outreach

Broad campaigns reached low-risk customers while high-risk SMBs slipped away unnoticed.

Real revenue at stake

SMB accounts carry high LTV — a single percentage point of churn translates into millions in foregone revenue.

Slow ML cycle time

Each new model took ~4 weeks to deploy, making it hard to respond as portfolio behavior shifted.

We aligned on a sharp, measurable definition before writing code.

THE QUESTION

Which active SMB cardholders are most likely to churn in the next 90 days — and what is the right action for each tier?

CHURN, DEFINED

No eligible card spend for 90 consecutive days after the as-of date.

Co-signed with Marketing & Finance

POPULATION

Active SMB cardholders, scored monthly with a 24-month look-back.

Snapshot-based labeling

PRIMARY METRIC

Incremental churn reduction vs. holdout — translated to retained revenue.

Causal, not correlational

SECONDARY METRICS

PR-AUC, lift @ top decile, calibration, and pipeline run-success rate.

Tech + business in one scorecard

Six stages, one accountable owner, fully productionized.

I owned the entire arc — from problem definition to live scoring and monitoring — partnering with Marketing, Finance, Risk, and Engineering at each stage.

STAGE 01	STAGE 02	STAGE 03	STAGE 04	STAGE 05	STAGE 06
Frame	**Source**	**Engineer**	**Model**	**Deploy**	**Monitor**
Define churn, population, KPIs	24 mo. of txn, account, engagement	120+ features across 6 families	Elastic-net logit vs. boosted-tree	CI/CD batch scoring pipeline	Drift, retraining, campaign feedback

WHY THIS MATTERS

A single owner across the full lifecycle eliminated handoff loss, kept the model interpretable for stakeholders, and turned the pipeline into a reusable template.

24 months of behavior, snapshotted with no leakage.

DATA SOURCES

Transactions

Spend, ticket size, MCC mix, channel (online vs. in-person), 24-month history

Account & product

Tenure, product tier, credit limit, autopay, segment, region

Engagement & risk

Marketing contacts and responses; payment status, delinquencies, fees

POINT-IN-TIME LABELING

LEAKAGE-PROOFED BY DESIGN

Features use only data observable up to T.

Labels use only data observed strictly after T.

Splits are time-based (older months train, newer months validate).

CLASS IMBALANCE · Churn is the minority class — handled with class weights and threshold tuning, not naive resampling, so calibrated probabilities remain meaningful.

120+ features, six families — every one tied to a hypothesis.

I started from business hypotheses ("why do SMBs stop using their cards?") and translated each into measurable signals across multiple time windows.

Recency & Frequency

Days since last txn, active months in last 3/6/12, dormant-to-active ratio.

Spend & Ticket

Monthly spend, txn counts, average ticket, online vs. in-person share, MCC mix.

Trend & Stability

Δ spend vs. prior 90d, slope over 6 mo., volatility, rolling-window ratios.

Account & Product

Tenure, product tier, credit limit band, region, industry/size segment.

Payment & Risk

Revolver vs. transactor, delinquency flags, over-limit events, fees.

Engagement

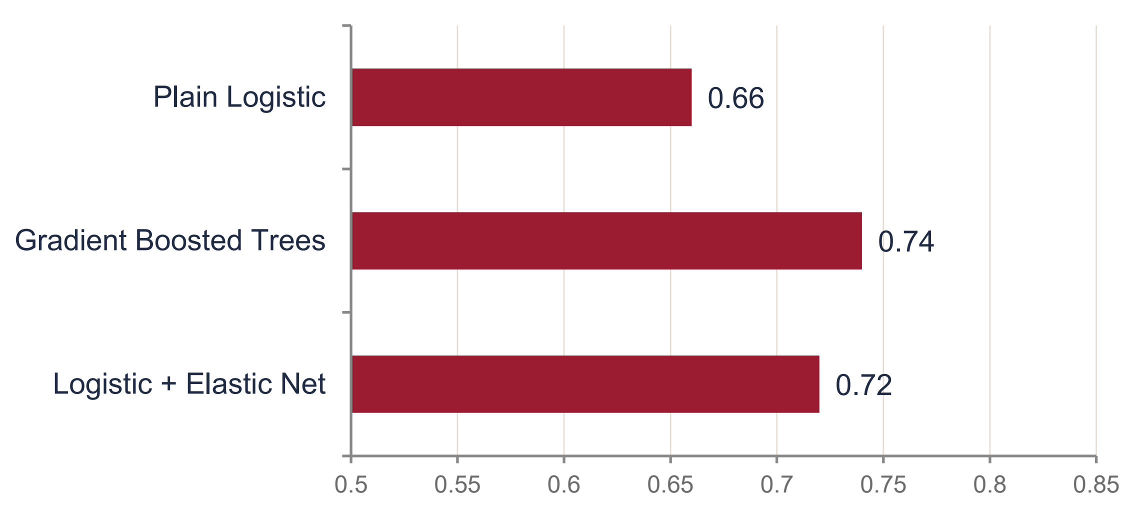
Past campaign contacts, response rates, channel preferences, recency of contact.

HOW WE NARROWED FROM CANDIDATES TO PRODUCTION SET

Quality filters › Leakage check › Correlation prune › Elastic-net selection › Stakeholder sanity check

Elastic-net logistic regression — performance close to GBM, far easier to defend.

BENCHMARK BAKE-OFF



GBM was marginally ahead. Elastic-net logistic was within striking distance — and won on interpretability, calibration, and operational simplicity.

WHY ELASTIC-NET LOGISTIC WON

Interpretable coefficients

Marketing and Risk could read the model and challenge it — buy-in came faster.

Handles correlated features

L1+L2 penalty stabilizes coefficients across the highly correlated RFM + trend features.

Built-in feature selection

Coefficients consistently shrunk to zero across CV folds were dropped — fewer brittle inputs.

Calibration-friendly

Sigmoid output maps cleanly to risk tiers; isotonic calibration completed the job.

PRINCIPLE · *Pick the simplest model that meets the bar. Complexity has to earn its way in.*

Time-aware CV, calibrated probabilities, business-grounded thresholds.

HYPERPARAMETER TUNING

HYPERPARAMETER	RANGE	STRATEGY
C (inverse regularization)	**0.01 → 10**	*log-grid*
l1_ratio (L1 vs. L2 mix)	**0.1 → 0.9**	*step 0.2*
class_weight	**balanced / custom**	*imbalance*
decision threshold	**tuned on lift**	*business*
calibration	**isotonic**	*post-fit*

Search via stratified, time-aware K-fold CV on PR-AUC. Selection favored configurations that were stable across folds, not just the single best.

VALIDATION & METRICS

Time-aware splits

Train on older months, validate on newer. No future peeking — production realism baked in.

Optimize PR-AUC + lift @ top-decile

AUC alone misleads on imbalanced labels. Lift ties metrics directly to campaign efficiency.

Probability calibration

Isotonic calibration ensures p = 0.7 actually means 70% — critical for risk tiering.

Subgroup stability

Performance audited across tenure, region, segment to surface drift or fairness issues.

OUTPUT · Calibrated probabilities mapped to High / Medium / Low risk tiers — each tier wired to a distinct marketing treatment.

From notebook to monthly scoring — automated, tested, monitored.

Extract
Scheduled pull from warehouse

→

Features
Versioned, reusable feature modules

→

Score
Versioned model produces probs + tiers

→

Persist
Scores written to Marketing tables / API

→

Monitor
Drift, freshness, score distributions

</> CI/CD that earned the win

Code in Git, unit tests on feature logic, integration tests on sample data, automated deploys on merge — eliminating four weeks of manual handoffs.

Production guardrails

Schema checks, null-rate alerts, score-distribution drift, feature drift, and pipeline run-success monitoring — 97% success across runs.

DEPLOYMENT TIME · 4 weeks → < 1 week | **PIPELINE SUCCESS RATE** · 97%

Scores became actions — every tier had a treatment.

HIGH RISK

Top decile

TREATMENT

Proactive retention

- Personalized outreach by relationship manager
- Targeted offer or fee-waiver
- Higher-touch cadence over 60 days

MEDIUM RISK

Next 2–3 deciles

TREATMENT

Nudge & educate

- Email and in-app reminders
- Education on under-used product features
- Light-touch incentive triggers

LOW RISK

Remainder

TREATMENT

Business-as-usual

- Standard servicing
- Cross-sell where relevant
- Avoid over-contact / opt-out fatigue

PRIORITIZATION · `priority = churn_probability × account_value` · *spend the next dollar where it saves the most.*

Holdouts proved the lift was real, not coincidence.

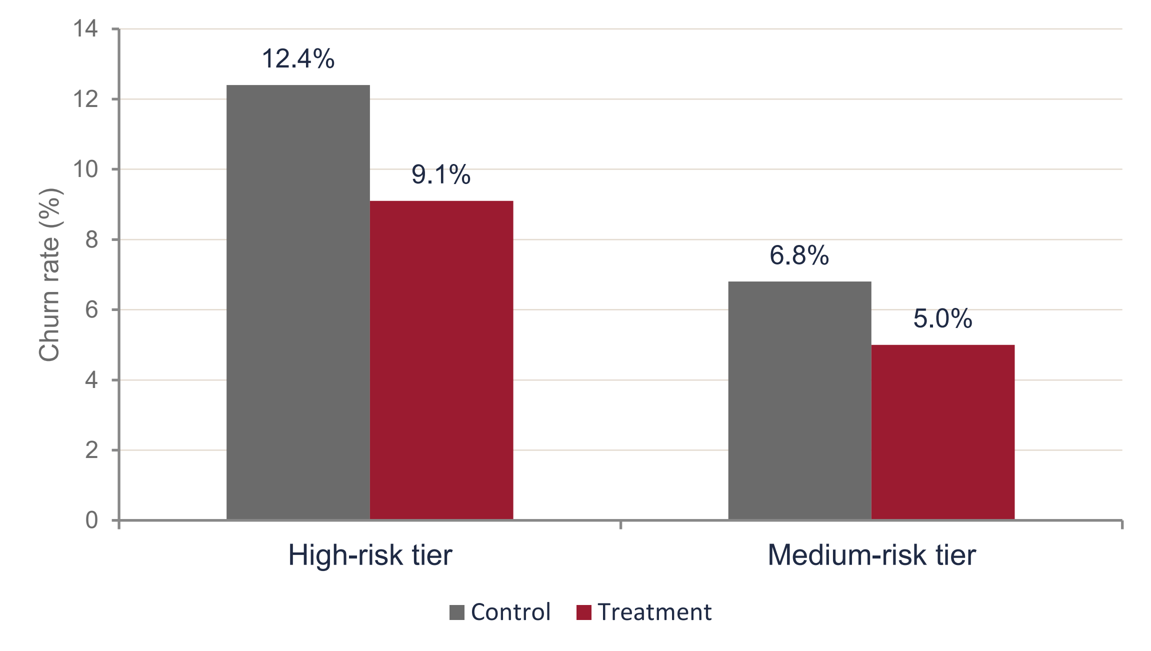
THE EXPERIMENT

We held back a randomized control group within high-risk and medium-risk tiers. Treated customers received targeted retention treatments; control received business-as-usual.

WHY A HOLDOUT MATTERED

Without a holdout, any drop in churn could be attributed to seasonality, macro shifts, or the campaign — we wouldn't know which. The holdout cleanly separates signal from noise and quantifies incrementality.

CHURN RATE — TREATMENT vs. CONTROL



Illustrative — directional values shown for confidentiality.

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FINANCE-VALIDATED IMPACT

2.5–3.5% absolute reduction in SMB churn, translated by Finance to **$8–12M** in incremental annual retained revenue.

The scoreboard.

ABSOLUTE CHURN REDUCTION

2.5–3.5%

in the SMB cardholder portfolio post-launch

ANNUAL RETAINED REVENUE

$8–12M

Finance-validated, incremental vs. holdout

MODEL DEPLOYMENT TIME

4 wks → <1 wk

via CI/CD automation and modular feature code

PIPELINE SUCCESS RATE

97%

monitored across monthly scoring runs

QUALITATIVE WINS

Churn scores adopted as a standard input into Marketing's campaign tooling · Pipeline reused as a template for downstream ML use cases · Cross-functional trust earned through interpretable model and clean measurement.

What worked, what I'd do differently next time.

WHAT WORKED

Frame the problem with stakeholders first

Aligning churn definition and success metrics with Marketing & Finance on day one prevented late-stage rework.

Pick the simplest defensible model

Elastic-net logistic bought us interpretability and stability without sacrificing meaningful performance.

Treat measurement as a first-class deliverable

The holdout design is the only reason we could credibly claim incremental revenue.

WHAT I'D DO DIFFERENTLY

Move from churn prediction to uplift modeling

Predicting who is likely to churn isn't the same as who is most persuadable — uplift directly optimizes the marketing dollar.

Close the loop with campaign response data

Feed past-treatment outcomes back into the next training cycle as features and labels.

Add automated retraining triggers

Move beyond cadence-based retraining to drift-triggered retraining for faster reaction to portfolio shifts.

A repeatable playbook, not a one-off project.

01 **Frame before you fit.**

Sharp churn definitions and finance-aligned success metrics turn ML output into business decisions.

02 **Interpretability is a feature.**

A model Marketing and Risk could read got adopted faster than a black-box ever would have.

03 **Engineering is half the win.**

CI/CD, monitoring, and modular features turned a one-shot model into a reusable platform.

Thank you · Questions?